

# केंद्रीय विद्यालय रानीखेत

Periodic Test-1

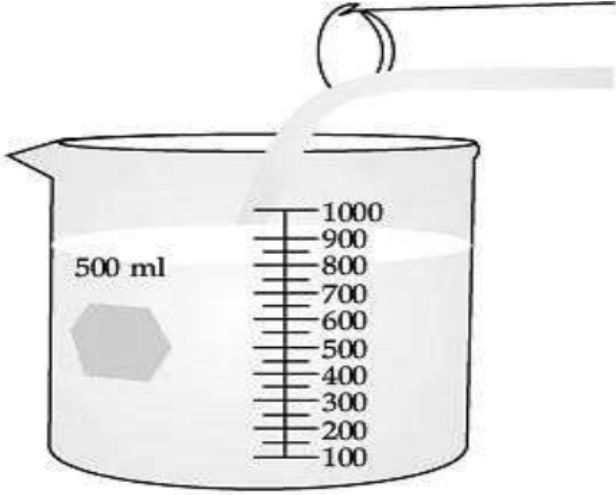
Class-11 Mathematics

TIME: 90 Min

M.M.40

**General Instructions:** All questions are compulsory.

| Q.No. | SECTION-A(1 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1.    | Find the number of the proper subset of {a, b, c}.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1     |
| 2.    | If $\left(\frac{x}{3} + 1, y - \frac{2}{3}\right) = \left(\frac{5}{3}, \frac{1}{3}\right)$ , then find the values of x and y.                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| 3.    | <p>a. Assertion is correct, reason is correct; reason is a correct explanation for assertion.</p> <p>b. Assertion is correct, reason is correct; reason is not a correct explanation for assertion</p> <p>c. Assertion is correct, reason is incorrect</p> <p>d. Assertion is incorrect, reason is correct</p> <p><b>Assertion:</b> cosec x is negative in third and fourth quadrants.</p> <p><b>Reason:</b> cot x decreases from 0 to <math>-\infty</math> in first quadrant and increases from 0 to <math>\infty</math> in third quadrant</p> | 1     |
| 4.    | The value of $(i^5)^{-6} + \frac{1}{i^{-17}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| 5.    | If $ x - 1  > 5$ , then find x.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| Q.No. | SECTION-B (2 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
| 6.    | If A and B are two sets such that $n(A) = 115$ , $n(B) = 326$ , $n(A - B) = 47$ , then find $n(A \cup B)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
| 7.    | Find the domain and range of the function $f(x) = \sqrt{25 - x^2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2     |
| 8.    | Find the angle in radian through which a pendulum swings if its length is 75 cm and tip describes an arc of length 21cm.                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| 9.    | If $(x + iy)(2 - 3i) = 4 + i$ , then find x and y.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2     |
| 10.   | Solve the inequality: $-(x - 3) + 4 > -2x + 5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2     |
| Q.No. | SECTION-C(3 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
| 11.   | Use the properties of sets to prove that for all the sets A and B, $A - (A \cap B) = A - B$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3     |
| 12.   | If $\tan x = \frac{3}{4}$ , $\pi < x < \frac{3\pi}{2}$ find the value of $\sin \frac{x}{2}$ , $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ .                                                                                                                                                                                                                                                                                                                                                                                                       | 3     |
| 13.   | If $\alpha$ and $\beta$ are different complex numbers with $ \beta  = 1$ , then find $\left  \frac{\beta - \alpha}{1 - \bar{\alpha}\beta} \right $ .                                                                                                                                                                                                                                                                                                                                                                                            | 3     |
| 14.   | Solve the inequality $\frac{2x-1}{3} \geq \frac{3x-1}{4} + \frac{x-1}{5}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3     |
| Q.No. | SECTION-D (5 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
| 15.   | Prove that $\sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos 8A}}} = 2 \cos A$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5     |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
|     | <p style="text-align: center;"><b>OR</b></p> <p>Prove that : <math>\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ = \frac{3}{16}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |
|     | <b>CASE STUDY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |
| 16. | <p>In a school, students of class XI were discussing about the relations and functions. Three Students Ankita, Babita and Kavita form three sets <math>A = \{1, 2, 3, 4, 5\}</math>, <math>B = \{2, 4, 6\}</math> and <math>K = \{5, 7\}</math>.</p>  <p>Based on the above information answer the following:</p> <p>(i) Find <math>n(A \times B)</math> and <math>n(K \times B)</math></p> <p>(ii) A correspondence of elements from A to B given as <math>\{(1, 2), (2, 2), (3, 4), (3, 6), (4, 4), (5, 6)\}</math>.<br/>Is it a function? Justify your answer.</p> <p>(iii) If the function <math>f: B \rightarrow K</math> such that <math>(b, k) \in f</math> and <math>b &lt; k</math>, defined by <math>f = \{(1, 5), (x, 5), (5, y), (2, 7), (4, 7)\}</math>, then find x and y.</p> | <p>1</p> <p>1</p> <p>2</p> |
| 17. | <p>A beaker contains 640 ml of 8% solution of boric acid. This is to be diluted by adding 2% solution of boric acid to it. Based on this information answer the following questions.</p>  <p>1. Find the initial quantity of water in the beaker.</p> <p>2. Find the initial quantity of Boric acid in the beaker.</p> <p>3. If x ml of 2% boric acid solution is added to the beaker, find the quantity of acid in the resulting solution.</p>                                                                                                                                                                                                                                                                                                                                             | <p>1</p> <p>1</p> <p>2</p> |